

Customer Retention & Revenue Risk Analytics for a Retail Bank

Project Overview

Customer Retention & Revenue Risk Analytics for a Retail Bank

Business Objective

Understand why customers are leaving, identify high-risk segments, and estimate the financial impact of churn.

Key Findings

- Global Churn Rate: 20.4%
- Total Deposits at Risk: \$185.68M
- High-Value Customers at Risk: 612 customers

Critical Insight: Germany's churn rate (32.4%) is nearly 2.0x France's, representing \$97.97M of deposits at risk and making it the bank's highest-priority retention market.

Executive Takeaway

Analysis of 10,000 customers revealed that inactive members and customers with unresolved complaints have significantly higher churn rates. The bank risks losing approximately \$185.68M in deposits if current trends continue. Targeted retention efforts on the German market and high-value inactive accounts could immediately protect millions in deposits.

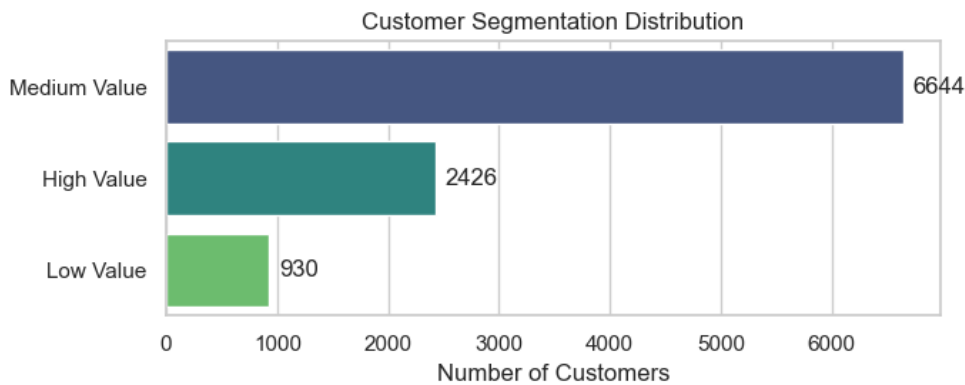
Project Outcome: This analysis identified three retention initiatives capable of protecting up to \$71.58M+ in customer deposits.

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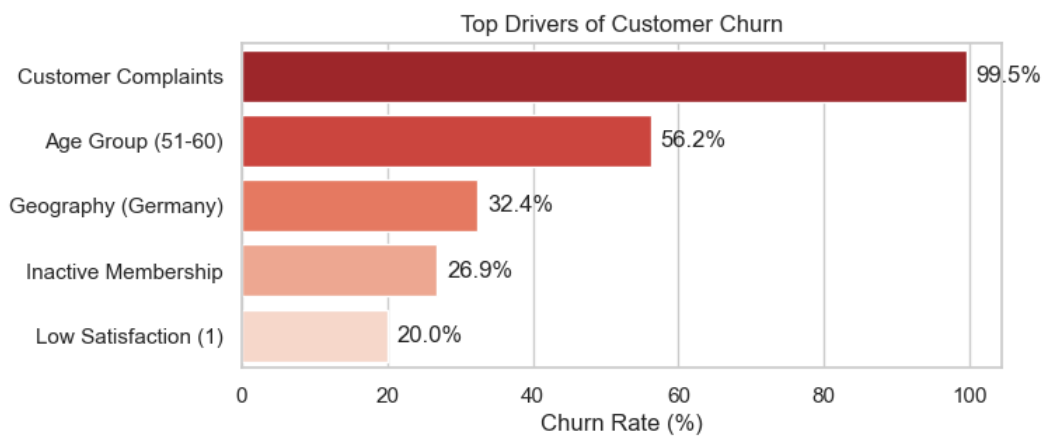
Key Insights

Insight 1: Customer Segmentation

- High Value Customers: 24.3% | Medium: 66.4% | Low: 9.3%

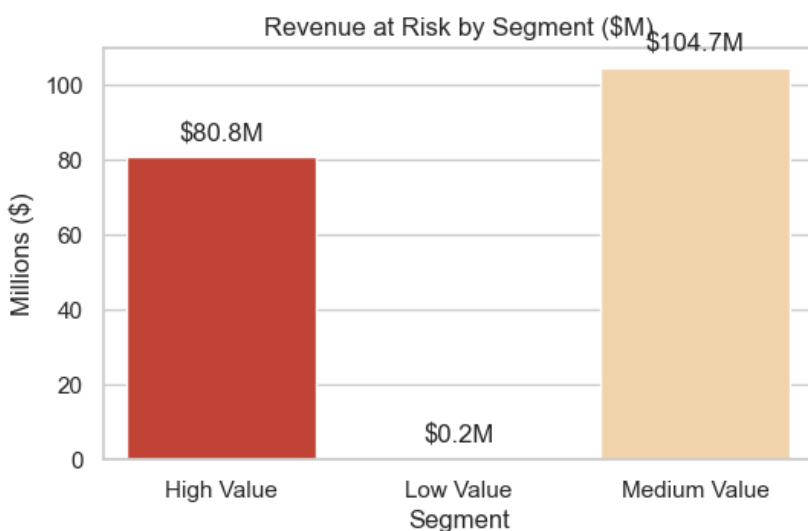


Insight 2: Churn Drivers



Insight 3: Revenue at Risk

- Total deposits currently at risk sum to \$185.68M.
Methodology: Revenue at Risk was calculated as the total balance held by churned customers.
- The breakdown by segment:



Recommendations & Business Impact

Recommendation 1: Retain High-Value Inactive Customers

- Why: Highest financial impact per customer.
- Potential Impact: Reducing churn in this specific segment by 10% would protect approximately \$5.15M in deposits.

Recommendation 2: Target High-Churn Regions (Germany)

- Why: Germany shows a severely elevated churn rate relative to its peers.
- Potential Impact: Normalizing Germany's churn rate to the global average would secure up to \$29.39M in at-risk funds.

Recommendation 3: Improve Complaint Resolution Pipelines

- Why: Strongest single relationship between an event (complaint) and churn execution.
- Potential Impact: Resolving 20% of high-risk complaints proactively could save \$37.04M in balances.

Methodology & Technical Depth

Key SQL Analyses Performed

- Customer Segmentation using CASE statements
- Revenue-at-Risk calculations aggregating across demographics
- Churn Ranking by Geography using Window Functions
- High-Risk Customer Identification using CTEs

Key Analytical Findings (Python/Pandas)

- Complaint registration showed the strongest relationship with churn (>95%).
- Inactive members were significantly more likely to leave than active members.
- Customer age positively correlated with churn probability (highest in 50-60 bracket).

Tools & Dataset

- Languages & Tools: SQL, Python (Pandas, Matplotlib/Seaborn), Power BI
- Dataset: Bank Customer Churn Dataset (10,000 records)

What Should The Bank Do Next?

Priority	Action	Expected Impact
High	Retain inactive high-value customers	Protect \$5.15M
High	Improve complaint resolution	Protect \$37.04M
Medium	Focus Germany retention campaign	Protect \$29.39M
Medium	Loyalty program for new customers	Improve retention